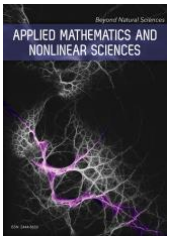
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A study on the application of game theory modeling in the analysis of pharmaceutical price elasticity

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Abstract

Setting reasonable drug prices is the fundamental goal of drug price elasticity analysis, and the game theory method can play a role in drug pricing by analyzing the behaviors of multiple game subjects to give a plan that meets the best interests of all parties. The optimal pricing strategies under different threshold conditions are summarized by establishing four model assumptions and solving them based on the basic model of game theory and the bidding process. Accordingly, the drug pricing of Pharmaceutical Company F was analyzed, and it was found that its optimal pricing was 21.3yuan, at which time the profit was 132.74% higher than the original pricing profit. After applying this pricing, it was found that the implementation of the new pricing increased the mean score of the company's profitability by 0.16738 to 0.19504 as compared to before the optimized drug pricing strategy, indicating that the pricing strategy improved by the game theory model is able to provide the company's level of profitability and is feasible.

Keywords: Drug pricing; Game theory; Profitability level; Price elasticity.
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1 Introduction

As a special commodity, pharmaceuticals have both economic and social attributes, which makes them different from the general goods on the market, but also increases the difficulty of the government to effectively regulate the price of drugs. In terms of its attributes, drug price regulation belongs to the category of social regulation, which has been one of the important contents of the government's regulatory function [1-2]. Along with China's economy from a production-oriented gradually transitioning to a consumption-oriented, the market mechanism also gradually penetrated the pharmaceutical industry. Although the government has taken a variety of drug price regulation measures, the drug price problem is still not properly resolved. Drug prices have become an important social issue affecting the public's daily life [3-5]. In the context of drug prices dominated by the government for a long time, with the deepening of the degree of its false high, the call for the abolition of drug government pricing is growing. For the consumer demand for drugs, the pharmaceutical industry has basically reached a consensus on its inelasticity, and this is the premise of much in-depth research and analysis [6-8]. In the context of the marketization of government public services, the abolition of government pricing of medicines, reduce the direct intervention of the government, and the use of market mechanisms to regulate the price of medicines have become increasingly important, but in the context of the market mechanism is yet to be perfected, the withdrawal of government control does not mean that the market for medicines is completely free to develop, and the qualitative understanding of the need for quantitative research to support and validate the qualitative understanding of the market [9-10]. Based on this background, the study of drug price elasticity analysis is of great practical significance [11].

In the field of distribution of benefits from pharmaceutical distribution, many scholars have applied game theory to conduct useful explorations. Literature [12] shows that price theory is a neoclassical microeconomic analysis method that simplifies the model to solve distributional problems through prices, and this approach has important implications for research in areas such as market design and international trade and requires further clarification of its accuracy and validity. Literature [13] proposed a price game model with heterogeneous expectations and found through empirical analysis that there are complexity and bullwhip effects among various channels under different price adjustment speeds and historical price discount sensitivity conditions and that controlling this factor of price discount sensitivity within an appropriate range allows firms to achieve stable operations and avoid unnecessary risks. Literature [14], after in-depth discussion and documentation of the differences in cost-sharing of different drugs in private drug plans and the association with price elasticity of demand, concluded that further attention to and optimization of these decision-making processes are necessary and that only through the scientific and rational setting of consumer cost-sharing, taking into account the interests of all parties and social well-being, can we achieve the goal of long-term sustainable development in the field of health care. Literature [15] constructs a sustainable closed-loop pharmaceutical supply chain system based on a scenario-based multi-objective mixed-integer linear programming model and introduces a scenario-based game theory model into the system, which is analyzed experimentally to show that the proposed system is of great significance in promoting the recycling of resources, meeting the needs of customers, and reducing the impacts on the environment, and provides a reference for the future development of the related fields. Literature [16] reveals that executives of a company that produces improved treatments for multiple sclerosis take into account a number of factors when setting the price of a product through a game mechanism and further explore how these factors are related to each other and have an impact on the outcome.

This paper introduces the basic theories of game theory, such as the three elements of game theory and Nash equilibrium, on the basis of which a game theory model is applied to analyze the game process of competitive bidding by two to three pharmaceutical companies. Four model assumptions

are developed based on the bidding rules, and then each assumption is validated. The validation process explores the relationship between the final drug pricing outcomes and preferences and the premium capacity of cost-advantaged drug companies, the external options and rate-of-return objectives of disadvantaged cost drug companies, and the amount of costs incurred by pharmaceutical companies. The conditions under which multiple drug companies adopt either a high-price profitability strategy or a low-price blocking strategy are summarized, and the situation in which the interests reach a Nash equilibrium under a multi-party game is calculated, according to which the probabilities of drug companies implementing different pricing strategies are estimated. Finally, this paper also analyzes the feasibility of the pricing strategy by taking Company F as an example, and uses the entropy weight method to design indicators to test the impact of the pricing strategy on the company's profitability performance.

2 Drug pricing strategies based on game theory

2.1 Game theory

Game theory, also known as response theory, is a mathematical theory and methodology for the study of phenomena of a competitive nature and is mainly used to analyze how individuals select options that maximize their interests when they produce interactions [17]. The modern game theory originated from Game Theory and Economic Behavior, published in 1944, and Tucker revealed the phenomenon of the Prisoner's Dilemma in 1950, which provided a prototype for the study of non-cooperative games. After that, John Nash proposed the famous Nash equilibrium theory, which successfully explains how the two sides of the game can reach the equilibrium state under the premise of possessing incomplete information in non-cooperative games, marking a new beginning in the development of game theory.

2.1.1 The three elements of the game

1) The Decision-making subject, the participants of the game, refers to the subject who has the right to decide the strategy and bear the final result in the action. The game process can make its gain through decision-making, and the finite rational individual will only choose the decision with the greatest gain. Generally speaking, there are at least 2 decision-making subjects participating in the game, and x is used to represent the set of decision-making subjects, $x=\{1,2,3,\ldots n\}$.

2) Strategy: the action plan available to the decision-making subject is called strategy. For each decision-making subject participating in the elbow grease, there are at least 2 strategies available to him. The strategy s_i of the decision-making subject i can be expressed as $s_i=\{a,b,c\}$.

3) The gain function refers to the gain obtained by the decision-making body after each decision-making body chooses a strategy in Boben. The gain function depends on each decision-making body, not a decision-making body that can decide when n decision-making body decides the time of $\{s_1,s_2,\ldots s_n\}$, the decision-making body gain function for the time of $\{F_1,F_2,\ldots F_n\}$.

2.1.2 Nash equilibrium

In general, game theory can be categorized into cooperative and non-cooperative games, and the difference between non-cooperative and cooperative games lies in the ability of decision-makers to reach an agreement and act among themselves. If it is reached, it is a cooperative game, and a cooperative game will result in an increase in the interests of at least one party without harming the other individuals. If it is not reached, it is a non-cooperative game, and all decision-makers will act on behalf of the interests of individuals. The Nash equilibrium is an important concept known as the non-cooperative game equilibrium [18].

A Nash equilibrium is a game in which, assuming that there is n decision maker involved in the game, regardless of the decisions of the rest of the decision-makers, the parties are bound to choose a certain determined strategy, which is called the dominant strategy. Suppose the current combination of strategies, each decision maker in the other decision maker's strategy has been determined, its choice of strategy is optimal. In that case, the strategy combination is called Nash equilibrium. Nash equilibrium is divided into "pure strategy Nash equilibrium" and "mixed strategy Nash equilibrium." Nash proved that there must be a Nash equilibrium if each decision maker has a finite number of strategies and mixed strategies are allowed. The following is a classic prisoner game to illustrate the Nash equilibrium:

Suppose two criminals are arrested after committing a crime together, and they are unable to communicate. The police interrogate the two criminals separately, and the two criminals can choose between a confession and a non-confession strategy; if both criminals choose to confess, then the result is that both will be sentenced to 5 years imprisonment. If both offenders choose not to confess, they will be sentenced to one year's imprisonment. If one confesses and the other does not, the confessing party will be sentenced to three months in prison, and the other party will be sentenced to 10 years in prison. This is the classic Prisoner's Dilemma, where two criminals will be put in the position of choosing whether or not to confess, and without knowing how the other will choose, both will choose to confess for their good.

In the above example, it is found that both parties do not take the option that can make both parties reach the optimum. That is, both parties do not confess. This shows that the Nash equilibrium in a non-cooperative game reacts to the fact that individual rationality may produce results that are detrimental to others.

2.2 Bidding game between two or three pharmaceutical companies

2.2.1 Problem description and model assumptions

In the case of a lower-than-the-ceiling price A set by the government procurement department, three pharmaceutical enterprises simultaneously conduct a quotation session for a certain drug. The procurement department takes the lowest price among the prices quoted by the three pharmaceutical enterprises as the winning price and then asks the other two pharmaceutical enterprises whether they accept the price; if they do, they are also awarded the bid. If they do not accept it, the bidding is aborted (not awarded). Purchasing volume is determined according to the number of winning enterprises: if only one pharmaceutical enterprise wins the bid, that enterprise is entitled to 50% of the purchasing volume of the government procurement department. If two pharmaceutical companies win the tender, the two winning pharmaceutical companies will equally share 60% of the procurement volume of the government procurement department. If all three pharmaceutical companies win the bid, the three winning pharmaceutical companies will share 70% of the procurement volume of the

government procurement department equally. Based on the above rules, the following four assumptions are made in this paper.

Assumption 1: According to the microeconomics price demand model $Q=f\left(P\right)$ definition, the price P for the monotonically decreasing function of demand, so set the procurement sector inverse demand function for $P=\left(A-Q\right)k$, that is, the demand function is $Q=A-kP$, $k>0$ [19]. When the exclusive pharmaceutical company wins the bid, the company can government procurement department procurement volume of 50%, that is:

$$Q_1=Q/2 \tag{1}$$

When two pharmaceutical companies win the bid, then the two winning pharmaceutical companies will equally share 60% of the purchases made by the government purchasing department, i.e.:

$$Q_2=q_1+q_2=0.6*Q \tag{2}$$

When all three pharmaceutical companies win the bid, then the three winning pharmaceutical companies will equally share 70% of the purchasing volume of the government procurement department, i.e:

$$Q_3=q_1+q_2+q_3=0.7*Q \tag{3}$$

Assumption 2: Because the model methodology and results are similar, so here we choose to take the maximum value of the three pharmaceutical companies as an example. Three oligopoly pharmaceutical companies' cost function is shared information with cost structure asymmetry. That is, one of the pharmaceutical companies (assumed to be the pharmaceutical company 1) has a cost advantage, its fixed unit cost of C_L, and the other two have a cost disadvantage, the fixed unit cost of C_H, and $C_L<C_H$.

Hypothesis 3: Leader-Follower Hypothesis of Bidding Offers: Pharmaceutical firm 1 with the lowest unit cost wins the bid, and the other two pharmaceutical firms choose to reject the offer and forfeit the bid or accept the offer and win the bid.

Hypothesis 4 (Incomplete Information Hypothesis): The pharmaceutical companies are unable to determine the selection strategies of their competitors and, therefore, cannot predict the eventual market demand they will face.

2.2.2 Model solving

When pharmaceutical firm 1 uses the circled market strategy, the bid is slightly lower than C_H, at which point pharmaceutical firm 1's profit is:

$$\begin{aligned}\pi_1&=\left(C_H-C_L\right)Q_1/2\\&=\frac{AC_H-kC_H^2-AC_L+kC_HC_L}{2}\end{aligned} \tag{4}$$

At this point, Pharmaceutical Firm 2 and Pharmaceutical Firm 3 refuse to follow up on Pharmaceutical Firm 1's offer because acceptance of the offer would result in negative profits, resulting in zero revenue for Pharmaceutical Firm 2 and Pharmaceutical Firm 3.

There are three possible scenarios of market capacity when pharmaceutical firm 1's offer is higher than the cost of pharmaceutical firm 2 and pharmaceutical firm 3.

When pharmaceutical company 1 offers a price slightly lower than C_H, the sign letters in the return matrix represent only positive and negative returns, and their specific data are not calculated. This is because the Nash equilibrium solution to the revenue matrix can be obtained by giving a qualitative analysis of the positive and negative dimensions of the revenue.

In this case, when Pharma Firm 1 offers C_H, Pharma Firm 2 and Pharma Firm 3 will make negative profits if they accept the pricing, so (accept, accept), (reject, accept), and (accept, reject) are all inferior strategies, and thus the Nash equilibrium is (reject, reject). There are three possible market structures when pharmaceutical firm 1 quotes a price higher than C_H, where Π_1 of equation (4) is the profit of the blocking strategy, and the other variables are calculated as follows:

$$\pi_2 = \frac{(P_H - C_L)(A - kP_H)}{2} \tag{5}$$

$$\pi_3 = 0.3(A - kP_H)(P_H - C_L) \tag{6}$$

$$\pi_4 = \frac{7(A - kP_H)(P_H - C_L)}{30} \tag{7}$$

$$\pi_5 = \frac{7(A - kP_H)(P_H - C_H)}{30} \tag{8}$$

$$\pi_6 = 0.3(A - kP_H)(P_H - C_H) \tag{9}$$

Due to the existence of certain external selection and profitability objectives, Pharmaceutical Firm 2 and Pharmaceutical Firm 3 may not accept Pharmaceutical Firm 1's offer when their profits are positive. Hence, Pharmaceutical Firm 1's expected return is:

$$\begin{aligned} E\pi &= P^2\pi_4 + 2P(1-P)\pi_3 + (1-P)^2\pi_2 \\ &= \left(\frac{2}{15}P^2 - \frac{2P}{5} + \frac{1}{2}\right)\left(-kP_H^2 + AP_H + kC_L P_H - AC_L\right) \\ &= \left(\frac{2}{15}P^2 - \frac{2P}{5} + \frac{1}{2}\right)(A - kP_H)(P_H - C_L) \end{aligned} \tag{10}$$

From Eq. (10), the first-order derivation of P_H when quoting P_H to maximize profit yields the first-order condition shown in Eq. (11):

$$\frac{d\pi}{dP_H} = \left(\frac{2}{15}P^2 - \frac{2P}{5} + \frac{1}{2}\right)(-2kP_H + A + kC_L) = 0 \tag{11}$$

Solution:

$$P_H = \frac{A + kC_L}{2k} \tag{12}$$

This leads to the maximum expected profit:

$$E\pi = \left(\frac{2}{15}P^2 - \frac{2P}{5} + \frac{1}{2}\right)\frac{\left(A - kC_L\right)^2}{4k} \tag{13}$$

2.2.3 Strategy Analysis

Assuming that pharmaceutical firm 1 is risk-neutral, whether pharmaceutical firm 1 adopts a market circling strategy with an offer price slightly below C_H or a high yield strategy with an offer price higher than C_H depends mainly on the comparison between the two of Π_1 in equation (4) and $E\Pi$ in equation (13). It is important to note that probability P is an important influence on the strategic choice of pharmaceutical firm 1, while it is exogenously determined by the external choices or yield targets of pharmaceutical firm 2 and pharmaceutical firm 3. Therefore, the following proposition is made:

For the sake of illustration, $\Delta\Pi = \Pi_1 - E\Pi$ is first defined as the pricing strategy selection tendency if $\Delta\Pi > 0$, a low price blocking strategy is adopted, if $\Delta\Pi < 0$, a high price profitability strategy is adopted, thus $\Delta\Pi$ changes in the situation determine what pricing approach will be adopted by pharmaceutical company 1.

Proposition 1: When $\dfrac{k\left(kC_L + A\right) - k\sqrt{\left(A - kC_L\right)^2(1+2F)}}{2k^2} < C_H \le \dfrac{k\left(kC_L + A\right) + k\sqrt{\left(A - kC_L\right)^2(1+2F)}}{2k^2}$, pharmaceutical drug company 1 (cost advantage drug company) should choose price blocking strategy, and when $C_H > \dfrac{k\left(kC_L + A\right) + k\sqrt{\left(A - kC_L\right)^2(1+2F)}}{2k^2}$ or $C_I < C_H < \dfrac{k\left(kC_I + A\right) - k\sqrt{\left(A - kC_L\right)^2(1+2F)}}{2k^2}$, pharmaceutical company 1 should choose high price profitability strategy.

Proof: according to $\Delta\Pi = \Pi_1 - E\Pi$, i.e.:

$$\begin{aligned}\Delta\Pi = &\frac{AC_H - kC_H^2 - AC_L + kC_HC_L}{2} \\ &-\left(\frac{2}{15}P^2 - \frac{2P}{5} + \frac{1}{2}\right)\frac{\left(A - kC_L\right)^2}{4k}\end{aligned} \tag{14}$$

So, the conditions of the low price blocking strategy as an example:

$$2k\left(A - kC_H\right)\left(C_H - C_L\right) \ge \left(\frac{2}{15}P^2 - \frac{2P}{5} + \frac{1}{2}\right)\left(A - kC_L\right)^2 \tag{15}$$

To wit:

$$\begin{aligned}&-2k^2C_H^2+2k\left(kC_L+A\right)C_H-2kAC_L\\&-\left(\frac{2}{15}P^2-\frac{2P}{5}+\frac{1}{2}\right)\left(A-kC_L\right)^2\geq 0\end{aligned}\tag{16}$$

For ease of computation, the combined probability coefficients $F=\left(\frac{2}{15}P^2-\frac{2}{5}P+\frac{1}{2}\right)$, $a=-2k^2$, $b=2k\left(kC_L+A\right)$, and $c=-2kAC_L-\left(\frac{2}{15}P^2-\frac{2P}{5}+\frac{1}{2}\right)\left(A-kC_L\right)^2$ are defined.

Since the second-order derivative $\frac{d^2\Delta\Pi}{dC_H^2}=-4k^2$ of Eq. (16) $\Delta\Pi$ is negative, the function is convex, and there is a maximum value. According to the discriminant law:

$$\Delta=b^2-4ac=4k^2\left[\left(kC_L+A\right)^2-4AkC_L+2F\left(A-kC_L\right)^2\right]=4k^2\left(A-kC_L\right)^2(1+2F)\geq 0\tag{17}$$

When $A\neq kC_L$, there exist two solutions $C_H=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$, i.e.:

$$\left\{\frac{-k\left(kC_L+A\right)\pm k\sqrt{\left(A-kC_L\right)^2(1+2F)}}{-2k^2}\right\}\tag{18}$$

So the following price strategy conclusion can be obtained when:

$$\frac{k\left(kC_L+A\right)-k\sqrt{\left(A-kC_L\right)^2(1+2F)}}{2k^2}<C_H\leq\frac{k\left(kC_L+A\right)+k\sqrt{\left(A-kC_L\right)^2(1+2F)}}{2k^2}\tag{19}$$

When pharmaceutical drug company 1 (cost advantage drug company) should choose a low-price blocking strategy.

When:

$$C_H>\frac{k\left(kC_L+A\right)+k\sqrt{\left(A-kC_L\right)^2(1+2F)}}{2k^2}\tag{20}$$

Or:

$$C_I<C_H<\frac{k\left(kC_L+A\right)-k\sqrt{\left(A-kC_L\right)^2(1+2F)}}{2k^2}\tag{21}$$

When it does, pharmaceutical firm 1 should choose a high-price profitability strategy.

Thus, the proof of Proposition 1 is complete.

Proposition 2: Other things being equal, the propensity of firm 1 to adopt a high-price profitability strategy increases and then decreases as the costs C_H of high-cost pharmaceutical firm 2 and pharmaceutical firm 3 increase.

Suppose we are pharmaceutical company 1, faced with potential competition or real division of the market by high-cost pharmaceutical companies 2 and 3. In that case, we need to analyze the costs of the other two pharmaceutical companies and use a blocking strategy until a threshold is reached. After a certain threshold is exceeded, the leader pricing strategy, i.e., the high-price profitability strategy, is used.

PROOF: Find the derivative with respect to C_H for equation (14) $\Delta\Pi$:

$$\frac{d\Delta\Pi}{dC_H} = \frac{A + kC_L}{2} - kC_H \tag{22}$$

Therefore, when $C_H \le \frac{A + kC_L}{2k}$, the derivative is greater than zero, and the outcome is cumulative, i.e., the profit margin is increasing, the stronger the attractiveness and personal desire. Conversely, the outcome is decreasing cumulatively, the profit differential decreases, and the weaker the attractiveness. Combining Proposition 1 and its proof, it can be seen that when $C_H \le \frac{A + kC_L}{2k}$, the stronger the propensity of drug company 1 in the corresponding strategy interval, while when $C_H > \frac{A + kC_L}{2k}$, the weaker the propensity of drug company 1 in the corresponding strategy interval.

Thus, Proposition II is proven.

Proposition 3: Other things being equal, as the top price A increases, if the combined probability is positive, the propensity of pharmaceutical firm 1 to adopt a high-price strategy decreases and then increases. If the combined probability is negative, the propensity of firms to adopt a low-price blocking strategy decreases and then increases.

PROOF: Find the derivative with respect to A for the pair of equations (14) $\Delta\Pi$:

$$\frac{d\Delta\Pi}{dA} = \frac{C_H - C_L}{2} - \left(\frac{2}{15}P^2 - \frac{2P}{5} + \frac{1}{2}\right)\frac{A - kC_L}{2k} \tag{23}$$

According to the combined probability coefficient $F = \left(\frac{2}{15}P^2 - \frac{2}{5}P + \frac{1}{2}\right)$, the tendency of pharmaceutical firm 1 to adopt a low-price blocking strategy will be enhanced when $A \le \frac{kC_H - kC_L + kFC_L}{F}$, and the tendency of pharmaceutical firm 1 to adopt a high-price profitability strategy will be enhanced when $A > \frac{kC_H - kC_L + kFC_L}{F}$. At this point, the proof of Proposition 3 is complete.

Proposition 4: Other things being equal, as the cost C_L of pharmaceutical firm 1 increases, the propensity of pharmaceutical firm 1 to adopt a low-price blocking strategy increases and then decreases when the combined probability is positive, and the propensity of pharmaceutical firm 1 to adopt a high-price strategy increases when the combined probability is negative.

If we are Pharma 1, we need to add a markup to the top price based on our costs, taking into account the probability of entry of each of the high-cost Pharma 2 and Pharma 3, as well as their share of the market. If the combined probability is positive, our cost markup is relatively low. We tend to use a low-price blocking strategy, and if the combined probability is negative, our cost markup is relatively high. We tend to use a high-price profitability strategy.

Proof: find the derivative with respect to C_L for $\Delta\Pi$:

$$\frac{d\Delta\Pi}{dC_L}=\frac{kC_H-A}{2}-\left(\frac{2}{15}P^2-\frac{2P}{5}+\frac{1}{2}\right)\frac{kC_L-A}{2} \tag{24}$$

Since $F>0$, the tendency of pharmaceutical firm 1 to adopt a low-price blocking strategy is enhanced when $C_L\le\frac{kC_H-A+FA}{kF}$, and the tendency of pharmaceutical firm 1 to adopt a high-price strategy is enhanced when $C_L>\frac{kC_H-A+FA}{kF}$. Thus, the proof of Proposition IV is complete.

Proposition 5: Other things being equal, the propensity of pharmaceutical firm 1 to choose a low-price blocking strategy increases as the probability of pharmaceutical firm 2 and pharmaceutical firm 3 following up on pharmaceutical firm 1's entry into the market based on external selection P increases.

If we were pharmaceutical firm 1, we would need to collect a large amount of market information, estimate competitors' operating costs, external options, and profitability goals, and make pricing decisions based on the likelihood of their entry into the market. As the likelihood of competitors entering the market rises, the profitability of our high-pricing strategy diminishes, and we are, therefore, more inclined to choose a low-price blocking strategy.

Proof: find the derivative with respect to P for $\Delta\Pi$:

$$\frac{d\Delta\Pi}{dC_L}=\left(-\frac{4}{15}P+\frac{2}{5}\right)\frac{\left(A-kC_L\right)^2}{4k} \tag{25}$$

Since $\frac{d\Delta\Pi}{dP}\ge 0$, the proof of Proposition V is complete.

In summary, in a multi-oligopoly bidding process, due to the cost structure as common knowledge and the "leader-follower" offer outcome resulting from the bidding non-exclusivity rule, the final offer price is related to the premium capacity of the cost-competitive pharmaceutical firm, the external choice and profitability objectives of the disadvantaged pharmaceutical firm, and the cost amount of the pharmaceutical firm, so that pharmaceutical firm 1 needs to make a pricing decision taking into account the above conditions. The final offer is related to the premium power of the cost-competitive pharmaceutical company, the external options and yield targets of the disadvantaged cost pharmaceutical company, and the amount of costs incurred by the pharmaceutical company, so that

the pharmaceutical company needs to make a pricing decision taking into account all of the above conditions.

3 Analysis of pharmaceutical pricing strategies - the case of Company F

3.1 Validation analysis of pricing strategies

To verify the feasibility of the bidding strategy proposed in the previous section and solve specific problems of the pharmaceutical company, this paper uses Company F as an example for application analysis. Substituting the stated conditions into propositions I to V, respectively, the relationship between the cost C_H and the threshold with the change of the value of k is shown in Figure 1.

It can be noticed that the results vary with the value of k. This implies that Company F also has to consider the value of k, i.e. the extent to which the price of the drug affects the quantity demanded, when choosing its strategy. In reality, the quantity demanded of the drug is more or less affected by the price.

1) As shown in Figure 1, if the value of k is about 1.7-2.5, $\frac{0.7A+0.7kC_L-\sqrt{0.7(0.7-F)(A-kC_L)^2}}{1.4k}<C_H<\frac{0.7A+0.7kC_L+\sqrt{0.7(0.7-F)(A-kC_L)^2}}{1.4k}$ can be satisfied, and the market blocking strategy can be adopted, and if the value of k is small or large, the market sharing strategy can be adopted. In the same way, as the analysis of the impact of the cost C_H of the counterparty on the strategy choice of Company F, when the value of k is above 2.5, it satisfies $C_H>\frac{kC_L+A}{2k}$, that is, the tendency to choose the market blocking strategy is getting smaller and smaller. The degree of influence of price on demand is often affected by brand stickiness and drug quality effects. In the traditional sales model, due to the influence of drug effect and marketing strategy, drug brand stickiness is very strong. With the improvement of drug technology, the stickiness of drug brands has been greatly reduced. However, in order to ensure that clinical usage habits will still have a certain brand stickiness, that is, the k value will be in the transition stage from low to high, that is, between 1.7-2.5. Company F should adopt a low-price blocking strategy for this volume procurement, substitute the conditions into Equation $P^*=\frac{A+kC_L}{2k}$ and maximize profits, and the best pricing can be known to be 21.3 yuan. The verification is as follows: Company F adopts a low-price blocking strategy, that is, the price is 21.3 yuan, the total demand is obtained, the total area is 48218cm², the total profit is 183247.54 yuan, which is 132.74% higher than the original pricing profit. Therefore, the best strategy of Company F is the low-price blocking strategy, which is the most appropriate at 21.3 yuan. The bid-winning rate of this price is extremely high, and the raw materials and production plans can be arranged according to the purchase agreement corresponding to this ranking. With the gradual reduction of the cost of companies K and W, which are cost-disadvantaged, the tendency of company F to choose the blocking strategy will decrease first and then increase.

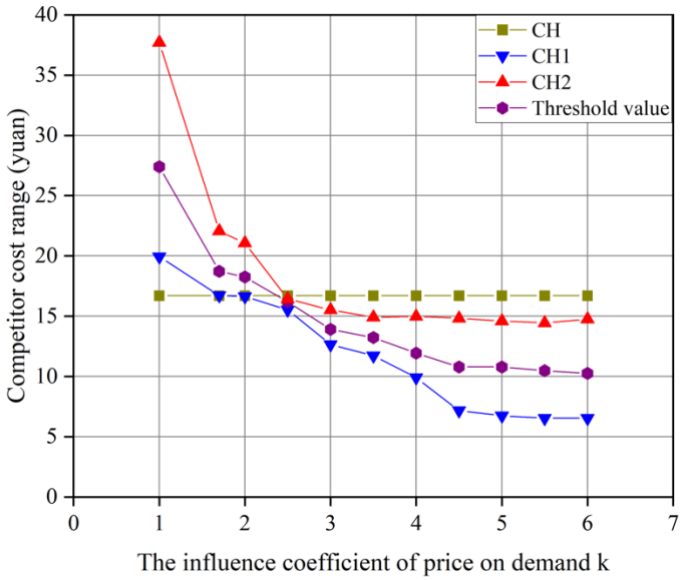


Figure 1. Relationship comparison of C_H and threshold value with k value

2) A comparison of the relationship between cost C_L and threshold k as a function of value is shown in Figure 2. Other things being equal, as the value of k increases, it satisfies $C_L \le \frac{1.4kC_H - 1.4A + AF}{Fk}$, i.e., the tendency to adopt the strategy of blocking the market increases as the cost of firm F grows. It is easy to understand that if only the degree of influence of price on the quantity demanded is taken into account, the lower the price, the greater the market share gained, as long as the cost is low enough to win the entire share by adopting the blocking market strategy. However, it should be noted that if the cost continues to increase, it is very likely to lose the cost advantage and the ability to block the market. Therefore, when Company F participates in the second band purchasing, if the cost of its products increases due to the increase in material costs, the tendency to choose the low price blocking strategy will first increase and then decrease.

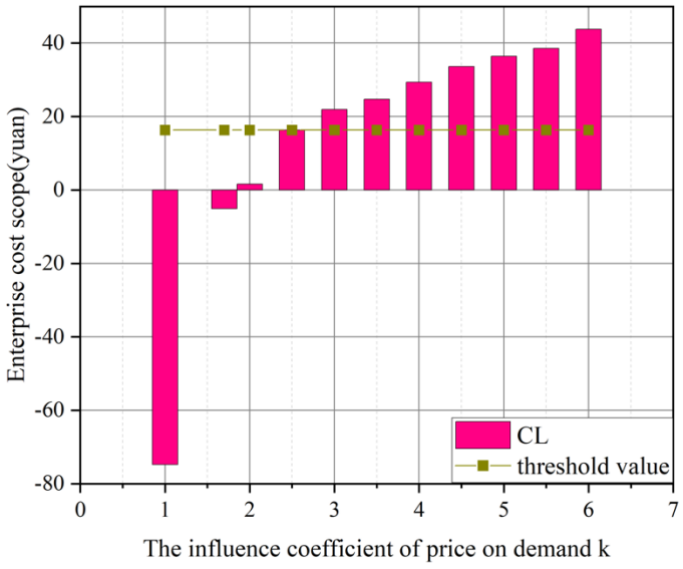


Figure 2. Relationship comparison of C_L and threshold value with k value

3) Fig. 3 shows the relationship between the top price A and the threshold as a function of the k value. As the value of k increases beyond 16.3, condition $A \leq \frac{1.4kC_H - 1.4kC_L + kFC_L}{F}$ is satisfied, i.e., as the value of A decreases, the strategy of adopting a blocked market is enhanced. In the actual situation, the top price is set lower and lower, which means that the profit space left for the enterprise is smaller. Even if it has a cost advantage, its single profit will not be very high. Then, the cost-advantage enterprise will naturally choose to block the market and win the maximum profit by retaining enough sales volume. Therefore, when Company F participates in the second band purchasing, the top price setting is lowered, and Company F prefers the low price blocking strategy.

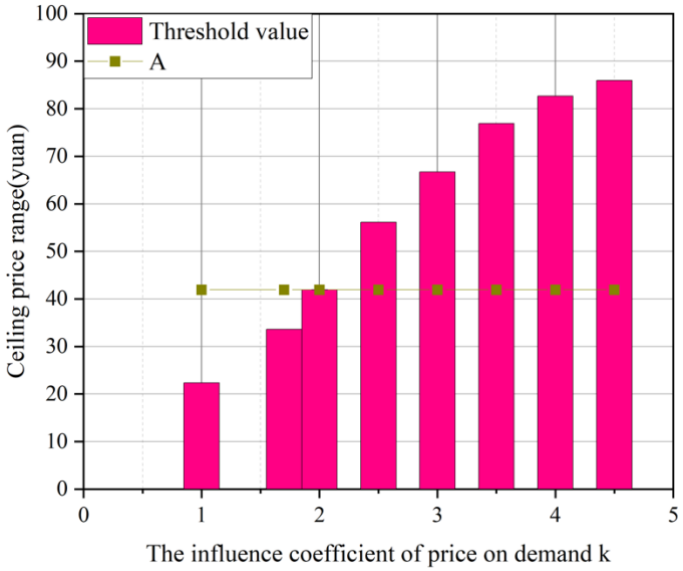


Figure 3. Relationship comparison of A and threshold value with k value

In summary, it can be seen that under this type of band purchasing rules, Company F, under the existing cost conditions, chooses the best low-price blocking strategy, and the probability of winning the bid is very high, according to which the procurement and production plan can be arranged in advance. Participating in the second round of banded purchasing can lead to choosing the low price blocking strategy as competitors' costs decrease. If your costs increase within a certain range, the tendency to choose the blocking strategy will increase first and then decrease. If the top price is reduced, it will be more likely to opt for the low-price blocking strategy. Therefore, in the case of even growth of the influence of price on the quantity demanded and the ability to control its costs, Company F still chooses the best low-price blocking strategy when participating in the second round of band purchasing.

3.2 Comparison of Profitability Performance of Firms

In order to verify how the pricing strategy derived based on the game model affects the profitability of Company F, this paper selects the entropy weight method to construct an index system to evaluate the profitability performance of Company F before and after the implementation of the new pricing strategy in July 2023.

3.2.1 Calculation steps of entropy weight method

1) Constructing a standardized matrix

When constructing the multi-indicator matrix, assuming that there are a evaluation object $(a1, a2, a3\ldots, am)$, b evaluation indicators $(b1, b2, b3, \ldots, bn)$, then define X_{ij} as the j th evaluation indicator $(i = 1, 2, 3, \ldots, m; j = 1, 2, 3, \ldots, n)$ of the i th evaluation object to build the evaluation matrix: $X = \{xij\} m \times n(i = 1, 2, 3, \ldots, m; j = 1, 2, 3, \ldots, n)$, the original matrix is:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}_{n \times m} \tag{26}$$

2) Data normalization processing

Since the indicators of different natures will have a large impact on the results, the influence of the scale should be removed in the calculation process, and the data should be normalized, and the processing is as follows:

The positive nature of the indicators is processed as follows:

$$X'_{ij} = \frac{X_{ij} - \min\{X_j\}}{\max\{X_j\} - \min\{X_j\}} \tag{27}$$

For negative indicators, the treatment is:

$$X'_{ij} = \frac{\max\{X_j\} - X_{ij}}{\max\{X_j\} - \min\{X_j\}} \tag{28}$$

The suitability indicator is treated as follows:

$$X'_{ij} = \frac{\max\{X_j\} - D_{best}}{\max\{X_j\} - D_{best}} \tag{29}$$

D_{best} is the standard value for determination

In order to ensure that the value is valid, each value is after the dimensionless addition of 0.0001 of the effective value.

3) Defining standardized values

Calculate the weight of the n nd indicator under each evaluation object as follows:

$$Y_{ij} = \frac{X'_{ij}}{\sum_{i=1}^{m} X'_{ij}} \tag{30}$$

4) Calculate the entropy value of indicators

The entropy value of each indicator is further calculated as follows:

$$e_j = -k\sum_{i=1}^{m}\left(Y_{ij} \times \ln Y_{ij}\right) \tag{31}$$

5) Calculate the degree of variation of the indicator

Calculate the degree of variability of the j st indicator as follows:

$$d_j = 1 - e_j \tag{32}$$

6) Calculation of indicator weights

Calculate the weight W_j of the j st indicator as follows:

$$W_i = d_j \Big/ \sum_{j=1}^{n} d_j \tag{33}$$

7) Calculation of single-indicator evaluation scores

The main purpose of this step is to be able to reduce the influence of subjective factors, using the weights calculated in the previous step and standardized data multiplied together:

$$S_{ij} = W_i \times X'_{ij} \tag{34}$$

3.2.2 F Analysis of the effect of the adjustment of the company's drug pricing strategy

1) Standardized results of financial data evaluation indicators

The results of the standardization of the financial performance evaluation indicators selected for F Pharmaceutical Company are shown in Table 1. It can be preliminarily seen that after the implementation of the new drug pricing strategy in July, the profitability of Company F improved significantly. For example, the net operating margin increased from 0.00367 to 0.32454, and the growth rate of operating income increased from 0.01263 to 0.37898.

Table 1. F Pharmaceutical company financial evaluation standardization results

Index	May	June	July	August	September	October
Quick ratio	0.20151	0.20926	0.13416	0.11585	0.21205	0.00875
Current ratio	0.0094	0.0086	0.0156	0.02204	0.01135	0.03896
Asset liability ratio	0.23086	0.237	0.20859	0.1566	0.08593	0.01236
Inventory turnover	0.85893	0.98998	0.72005	0.6569	0.36269	0.00179
Turnover of total assets	0.92806	1.00403	0.43832	0.66887	0.16707	0.02335
Accounts receivable turnover rate	0.62093	0.99917	0.52497	0.38916	0.10047	0.00495
Net operating margin	0.10916	0.00367	0.32454	0.59438	0.84246	0.98618
Net interest rate on total assets	0.15827	0.00764	0.34726	0.7148	0.85778	0.99686
Return on equity	0.25185	0.01997	0.46836	0.85943	1.00212	0.95698
Growth rate of total assets	0.19237	0.00703	0.6366	1.01037	0.58104	0.77672
Revenue growth rate	0.30024	0.01263	0.37898	0.97935	0.50577	0.14779
Growth rate of net assets	0.3151	0.00265	0.43282	1.00538	0.24468	0.24986

2) Construct weighting matrix

The standardized results of the above indicators of profitability are used to construct the weighting matrix, and the formula is used to calculate the entropy value and weight. The results of the entropy assignment of each indicator of profitability performance evaluation in 2023 of F Pharmaceutical Company are shown in Table 2.

Table 2. F Pharmaceutical company profit index entropy weight assignment

Primary index	Secondary index	Entropy value	Weight
Profitability	Net profit margin on sales	0.79391	0.10382
	Net interest rate on total assets	0.80786	0.09833
	Return on equity	0.83095	0.07332

3) Profitability Score

In order to provide a more precise and reliable analysis of F Pharma's profitability performance for six months, we can calculate the single-indicator evaluation scores for each of the profitability indicators based on the indicator weights derived in the previous section and rank them in order of ranking. From May to October, Pharmaceutical Company F's profitability scores are shown in Table 3.

Based on the calculation of the profitability scores in Table 3, it can be found that before applying the game theory model for drug pricing in July 2023, the profitability performance score of Pharmaceutical Company F was relatively stable, and the profitability performance score was maintained around 0.005. Since the use of a game theory model for drug pricing in July 2023, the company's profitability performance score has significantly increased, improving to 0.0932, an improvement of 87.02%. After that, the profitability performance score of Company F increased from August to October, and finally, the profitability performance score of the firm increased to 0.23644 one quarter after applying the game theory pricing model, which is an improvement of 0.23098 points from May. It shows that the game theory model helps improve the profitability of the company by helping to determine the price of drugs.

Table 3. Profitability score table for F company from May to October

Month		May	June	**July**	August	September	October
Profitability	Net profit margin on sales	0.01395	0.01761	0.04779	0.05491	0.10206	0.0845
	Net profit margin on total assets	0.00192	0.00512	0.05361	0.05085	0.09948	0.06846
	Net assets income rate	0.0015	0.0207	0.04975	0.05193	0.06928	0.07464
Profitability performance score		0.00546	0.04986	0.09325	0.19029	0.26019	0.23644

4 Conclusion

Taking Company F as an example, the relationship between cost and threshold is found the value of k is about 1.7-2.5, and Company F should adopt a low-price blocking strategy to maximize its profits. At this time, the drug pricing should be 21.3 yuan, and the theoretical total profit should be 183247.54 yuan, which is 132.74% higher than the original pricing profit. After Company F implements this pricing strategy, the effectiveness of the pricing strategy is verified by examining the profitability of Company F. The results show that the average profitability performance score of Company F was

0.02766 in the two months before the implementation of the new pricing strategy in July 2023, while the profitability score had reached 0.09325 in July, and the average profitability score in the next four months was 0.19504, which is a significant improvement from the profitability of Company F when the new pricing strategy was not implemented.

It shows that the research path of this paper to analyze and solve the pricing strategy of drug prices based on game theory is feasible, and the new drug pricing strategy is helpful in improving the profitability of pharmaceutical companies.

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